Unsupervised Classification – Lab Activity

Task 1: Perform Unsupervised Classification

Using the provided satellite image:

1. Perform **Unsupervised Classification** using **both**:
    - **K-Means algorithm**
    - **ISODATA algorithm**
2. After running both classifications, **write a short comparison** in your own words describing:
    - How K-Means works
    - How ISODATA works
    - The key differences between the two algorithms

Task 2: Reclassification of Classes

After completing the unsupervised classification outputs:

1. Examine the generated classes carefully.
2. Based on spectral tone, NDVI understanding, and visual interpretation, **merge/reclassify** the classes into the following **five major land-cover categories**:
    - Dense Vegetation
    - Grassland
    - Barren Land
    - Water
    - Built-Up Area
3. Export the final reclassified map and include it in your results.

Learning Resources

Use the following tutorials to understand the concepts of supervised and unsupervised classification.

https://www.youtube.com/watch?v=2SjzJHg4wIY

https://youtu.be/g_V6BVRz82E